

**7. GOVERNANCE & RELEASE GATE** · STPA Hazard Mitigation · Bumpless Override · Defensible Release Case

**1. SENSING**

Photons / Voltages to DMA

MIPI CSI-2 · SPI Bus Priority

**2. PERCEPTION**

ViT Encoders · DINOv2

3D Spatial Affordance Tokens

**3. MEMORY**

Temporal Belief ·  $SE(3)$  Trees

JEPA / RSSM World Models

**5. PLANNING (System 1.5 · MPU/NPU)**

Diffusion Policies · ACT Action Chunking

$H$ -Step Action Horizons · Jerk Continuity

**4. REASONING (System 2 · MPU)**

Vision-Language Foundation Models · 3D Goals

Leases with Expiring TTL ( $t_{\text{expire}}$ )

THE PROPOSAL-PERMISSION PRIVILEGE BOUNDARY

**6. REFLEX (System 1 · MCU)**

1 kHz Real-Time Loop · Control Barrier Functions ( $h(x) \geq 0$ ) · Stopping Distance  $d_{\text{stop}} \leq d_{\text{clearance}} \cdot \text{Veto}(u_t)$

**THE PHYSICAL WORLD** ( $W_t \rightarrow W_{t+1}$ ) · Kinetic Momentum · Joule Heat · Friction · Collision Dynamics

Endogenous Sensory Shift ( $O_{t+1}$ )